PROJECT-BASED LEARNING

PHASE-I EVALUATION

Medicine Reminder and Care Tracking System

Team ID: T125

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T125
Semester: 3rd
Section: A,E,ML1
Domain: *Health*

Team Members

1. *Aradhya Uniyal – 2027642*
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Mentor

Mr. Utkarsh Pant Sir

Interactions completed: 2

Problem Statement & Motivation

Problem Statement

Patients may forget medicines or care-related tasks(especially of old aged group who lives alone), while caretakers may struggle to monitor them consistently. Existing methods often rely on scattered or manual records. Our project addresses this gap by providing a centralized system to manage doctor-provided care plans, track medicine/task completion, maintain history, and generate summary reports for Assistants and patients.

Motivation

- *No connection or guidance after Checkup*
- *Patient doesn't remain regular*
- *Existing method can only give reminder but no records*

Target Users / Stakeholders

- *Patients and Assistants*
- *Patient's family members*
- *Easy care plan tracking*

Objectives

Project Objectives

1. *Develop a role-based medicine and care tracking system for patients and assistants.*
2. *Implement core functionalities for managing doctor-provided medicines, care tasks, schedules, and patient activities.*
3. *Integrate OOP and DSA concepts such as classes, inheritance, polymorphism, linked lists, queues, stacks, searching, and sorting.*
4. *Demonstrate practical applicability through adherence tracking, caretaker monitoring, record management, and summary reports, with scope for future enhancements.*

Key Objective: *Develop a system for patients to follow care plans, assistants to monitor activities, and generate organized progress reports.*

Proposed Solution

Proposed Approach

1. *Patient profile, medicines, tasks health records.*
2. *Scheduling, tracking, searching sorting.*
3. *Role-based Assistant/Patient workflow status tracking.*
4. *Store care plans, history health records using files.*
5. *Daily schedule, progress summary adherence reports.*
6. *Bridges Doctor's Instructions with Patient's Daily Care.*

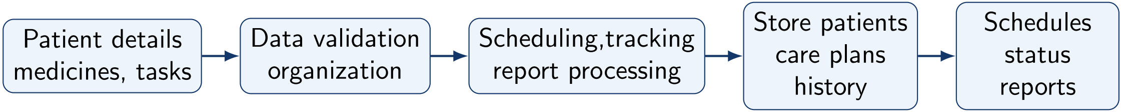
Key Features

- *Medicine Task Management*
- *Daily Reminders Schedule*
- *Taken/Skipped Task Completion Tracking*
- *History Health Record Tracking*
- *Progress Summary Report Generation*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Stacks, Queues, Searching, Sorting, Dynamic Memory Allocatio, Trees, Hashing*	*Patient Care Plan, Daily Schedule, Activity History, Search and Sorting,Patient and Medicine Lookup, Date/ID Based Record Management*
OOPs with C++	*Classes and Objects, Encapsulation, Abstraction, Inheritance, Polymorphism, Function Overloading/Overriding, Constructors, Destructors, Composition, Pointers, STL*	*Patient Management, Medicine Management, Task Management, Care Plan, Activity Records, Reports*

System Architecture / Workflow



Major Components

- *Authentication and Role Management*
- *Care Plan and Medicine*
- *History, Health Records and Reports*

Data / Control Flow

- *Assistant enters doctor provided instructions.*
- *System creates schedule and tracks patient actions.*
- *Reports show adherence, tasks health records.*

Technology Stack & Methodology

Technology Stack

- Programming: *C / C++*
- Database: *File Based Storage*
- Tools / Framework: *C++ Standard Library (STL)*
- IDE: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System and Module Design
3. OOP DS Implementation
4. Testing
5. Evaluation

C++ is selected for OOP and DS implementation with ALgorithms, while file-based storage provides simple persistent data management. STL supports standard programming utilities, and Git/GitHub enables version control and collaborative development.

Initial Progress & Team Contribution

Progress Achieved

- *Project topic and objectives finalized*
- *Requirements and user roles identified*
- *System modules and workflow designed*
- *Few modules implemented*

Individual Contribution

Member	Role	Contribution
M1	*Lead*	*Testing plan, documentation and presentation*
M2	*Developer*	*OOP class and module design*
M3	*DSA Developer*	*Data structures and algorithm design*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Requirement Analysis and System Design
2. Phase-II: OOP, DSA and Module Development
3. Phase-III: Integration, Testing and Final Implementation

Expected Outcomes

- *Working Medicine and Care Tracking System*
- *Daily Medicine/Task Tracking and History*
- *Supports patient care and adherence tracking*
- *Can be Used as a online platform*

References

1. *Data Structures and Algorithms book by G.S. Baluja*
2. *OOPs book by E Balagurusamy*
3. *C++ File Handling and STL Documentation*

Thank You
Questions & Suggestions